

THE GENERALIZED SECANT HYPERBOLIC DISTRIBUTION AND ITS PROPERTIES

David C. Vaughan

To cite this article: David C. Vaughan (2002) THE GENERALIZED SECANT HYPERBOLIC DISTRIBUTION AND ITS PROPERTIES, Communications in Statistics - Theory and Methods, 31:2, 219-238, DOI: [10.1081/STA-120002647](https://doi.org/10.1081/STA-120002647)

To link to this article: <https://doi.org/10.1081/STA-120002647>



Published online: 02 Sep 2006.



Submit your article to this journal [↗](#)



Article views: 477



View related articles [↗](#)



Citing articles: 5 View citing articles [↗](#)

DISTRIBUTIONS AND PROPERTIES

**THE GENERALIZED SECANT
HYPERBOLIC DISTRIBUTION
AND ITS PROPERTIES**

David C. Vaughan

Department of Mathematics, Wilfrid Laurier
University, Waterloo, Ontario, Canada N2L 3C5
E-mail: dvaughan@wlu.ca

ABSTRACT

The properties of a family of distributions generalizing the secant hyperbolic are developed. This family consists of symmetric distributions, with kurtosis ranging from 1.8 to infinity, and includes the logistic as a special case, the uniform as a limiting case, and closely approximates the normal and Student's t -distributions with corresponding kurtosis. A significant difference between this family and Student's t is that for any member of the generalized secant hyperbolic family, all moments are finite. Further, technical difficulties associated with evaluating moments of Student's t (especially for fractional degrees of freedom) are not present with this family. The properties of the maximum likelihood and modified maximum likelihood estimates of the location and scale parameters for complete samples are considered. Examples illustrate the methods developed in this work.

Key Words: Non-normality; Model selection; Parameter estimation; Modified maximum likelihood

1. INTRODUCTION

There are a number of situations where it is reasonable to assume that a random variable X has a symmetric distribution, but that distribution need not be normal. Praetz (1) and Blattberg and Gonedes (2) consider rates of return on common stocks, and McLeay (3) considers certain business related ratios, all finding Student's t with small degrees of freedom more appropriate than the normal, because of the kurtosis or thickness of the tails. Others such as Rocke et al. (4) and Spjøtvoll and Aastveit (5), in studies of grain yields, find there are reasons to consider distributions with tails thicker than the normal.

As suggested by the above, one of the more common alternatives to normality is Student's t -distribution. See, for example, Tiku and Suresh (6) and Bian and Tiku (7), who consider a scaled Student's t . As usual, the random variable U is said to have a Student's t -distribution with parameter $p \geq 1$ if its density function is given by:

$$f(u) = B^{-1}(1/2, (p-1)/2)(2p-1)^{-1/2}(1+u^2/(2p-1))^{-p} \quad (1.1)$$

where $-\infty < u < \infty$, and $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. We say U has $v = 2p - 1$ degrees of freedom. There are, however, a number of drawbacks to using this family: First, the k th moments do not exist for $k > 2p - 2$. In practice, using such a distribution as a model may not be convincing because of this, as noted by Spjøtvoll (8). The calculation of the moments of order statistics is difficult, even for a whole number of degrees of freedom, making the evaluation of typical parameter estimates, such as Lloyd's (9) BLUEs or Tiku's MMLs (Tiku and Suresh (6)), cumbersome. Barnett (10) has illustrated that there can be difficulties in calculating maximum likelihood estimates from thick tailed distributions. Further, Student's t does not allow for thin tailed distributions. Non-parametric methods can be very inefficient since they discard much of the structure inherent in the data. If a robust method of estimation is used, then at least 20% of the data may have to be discarded. See Tiku et al. (11), Chapter 4 for a discussion of censoring schemes that are appropriate for different alternatives to normality.

The purpose of this work is to introduce a family of distributions that eliminate many of these problems. Specifically, we will say Y has a standard *generalized secant hyperbolic* (GSH) distribution with parameter t if its density is given by:

$$g(y) = c_1 \frac{\exp(c_2 y)}{\exp(2c_2 y) + 2a \exp(c_2 y) + 1} \quad (1.2)$$



where for $-\pi < t \leq 0$:

$$a = \cos(t), \quad c_2 = \sqrt{(\pi^2 - t^2)/3} \quad \text{and} \quad c_1 = \frac{\sin t}{t} c_2 \quad (1.3)$$

and for $t > 0$:

$$a = \cosh(t), \quad c_2 = \sqrt{(\pi^2 + t^2)/3} \quad \text{and} \quad c_1 = \frac{\sinh t}{t} c_2. \quad (1.4)$$

We write $Y \sim \text{GSH}(0, 1; t)$, since it is simpler to express many properties in terms of t . This form is chosen so that $E(Y) = 0$ and $\text{Var}(Y) = 1$. Note also that regardless of the value of t , this distribution is symmetric about 0. The cumulative distribution function is:

$$\begin{aligned} G(y) &= 1 + \frac{1}{t} \cot^{-1}(-(\exp(c_2 y) + \cos t)/\sin t), & -\pi < t < 0 \\ &= \exp(\pi y/\sqrt{3})/(1 + \exp(\pi y/\sqrt{3})), & t = 0 \\ &= 1 - \frac{1}{t} \coth^{-1}((\exp(c_2 y) + \cosh t)/\sinh t), & t > 0. \end{aligned} \quad (1.5)$$

From the c.d.f., percentage points are easy to compute. Let $0 < u < 1$ and set $u = G(y)$. Then:

$$\begin{aligned} y &= \frac{1}{c_2} \ln[\sin(tu)/\sin(t(1-u))], & -\pi < t < 0 \\ &= \frac{\sqrt{3}}{\pi} \ln(u/(1-u)), & t = 0 \\ &= \frac{1}{c_2} \ln[\sinh(tu)/\sinh(t(1-u))], & t > 0. \end{aligned} \quad (1.6)$$

The moment generating function is defined, and takes the form:

$$\begin{aligned} M_Y(u) &= \frac{\pi}{t} \sin(ut/c_2) \csc(u\pi/c_2), & -\pi < t < 0 \\ &= \sqrt{3}u \csc(\sqrt{3}u), & t = 0 \\ &= \frac{\pi}{t} \sinh(ut/c_2) \csc(u\pi/c_2), & t > 0. \end{aligned} \quad (1.7)$$

Talacko (12) gives the characteristic function, using a somewhat different form of the constants. See also Johnson et al. (13), p. 143–148. A more general class of distributions was considered by Perks (14) in the context of graduating life-table data. Talacko (12) considered this specific family in the context of Weiner's stochastic variables.

Our goal is to introduce the GSH distribution in problems involving estimation of location and scale parameters when the underlying



distribution may be assumed symmetric. Tiku and Suresh (6) developed estimation methods for Student's t and other location-scale families, and Vaughan (15) shows censoring may be necessary for the Tiku-Suresh estimates to be defined. The GSH and Student's t with the same kurtosis are similar in structure but with some important differences. For example, the GSH family also includes thin-tailed members, with the uniform a limiting case, expanding the types of distributions that can be considered. Because of this, the GSH can be used effectively in modelling many different types of data.

2. PROPERTIES OF THE GSH

First note that when $t = -\pi/2$ then

$$g(y) = \frac{\exp(\pi y/2)}{\exp(\pi y) + 1} \quad (2.1)$$

the standard secant hyperbolic (whence the name). A generalization by Harkness and Harkness (16) involves powers of the secant hyperbolic and has been called a generalized hyperbolic secant. We note also that when $t=0$, the distribution is the standard logistic, which Tiku and Jones (17) show to be very close to the (standardized) t -distribution with 9 degrees of freedom. We also note that as $t \rightarrow \infty$, the density tends to that of the standard uniform distribution on $(-\sqrt{3}, \sqrt{3})$. The moment generating function satisfies:

$$\begin{aligned} M_Y(u) &= 1 + \frac{1}{2}u^2 + \frac{1}{4!} \frac{21\pi^2 - 9t^2}{5\pi^2 - 5t^2} u^4 + O(u^5), & -\pi < t < 0 \\ &= 1 + \frac{1}{2}u^2 + \frac{21}{5 \times 4!} u^4 + O(u^5), & t = 0 \\ &= 1 + \frac{1}{2}u^2 + \frac{1}{4!} \frac{21\pi^2 - 9t^2}{5\pi^2 - 5t^2} u^4 + O(u^5), & t > 0 \end{aligned}$$

establishing that $\text{Var}(Y) = 1$ and that the usual coefficient of kurtosis $\beta_2 (= E(Y^4) \text{ here})$ is

$$\begin{aligned} \beta_2 &= (21\pi^2 - 9t^2)/(5\pi^2 - 5t^2), & -\pi < t \leq 0 \\ &= (21\pi^2 + 9t^2)/(5\pi^2 + 5t^2), & t > 0. \end{aligned} \quad (2.2)$$



It is readily apparent that β_2 decreases as t tends to ∞ , and that $1.8 < \beta_2 < \infty$. Also,

$$t = -\pi \sqrt{\frac{5\beta_2 - 21}{5\beta_2 - 9}} \quad \text{when } \beta_2 \geq 4.2 \quad (2.3)$$

and

$$t = \pi \sqrt{\frac{21 - 5\beta_2}{5\beta_2 - 9}} \quad \text{when } \beta_2 \leq 4.2. \quad (2.4)$$

There is a unique member of the GSH family that corresponds to any given kurtosis for regular unimodal distributions: see Stuart and Ord (18), Exercise 3.18. In particular, when $t = \pi$ then $\beta_2 = 3$, the kurtosis of the normal. To illustrate the level of agreement between the member of the GSH family and corresponding standardized Student's t (i.e., with mean 0 and variance 1, when these moments exist), consider Table 1.

T_v is the Student's t variable with v degrees of freedom. This level of agreement implies that if a distribution truly is Student's t but we assumed the corresponding GSH with the same kurtosis, there would only be a small loss (if any) of efficiency. Note that if v is the value of the degrees of freedom for a t -distribution with a given (finite) kurtosis, then the parameter t in the GSH family is: if $4 < v < 9$ then $t = -\pi\sqrt{(9-v)/(v+1)}$; if $v=9$ then $t=0$; and if $v > 9$ then $t = \pi\sqrt{(v-9)/(v+1)}$. One reason for choosing this particular comparison of Student's t (scaled so the variance is 1) and corresponding GSH family member with the same kurtosis is that the distributions will have the same first four moments. Student's t with these moments is a particular case of the Pearson system of distributions

Table 1. Upper $\alpha \times 100$ Percentage Points of Selected Student's t (T_v) and GSH (Y) Variables with Common Kurtosis

β_2	$\alpha =$	0.001	0.01	0.025	0.05	0.10
9	Y	5.12	2.96	2.13	1.55	1.02
	T_5	4.57	3.12	1.99	1.56	1.14
4.2	Y	3.81	2.53	2.02	1.62	1.21
	T_9	3.79	2.49	2.00	1.62	1.22
42/13	Y	3.34	2.35	1.94	1.62	1.26
	T_{30}	3.27	2.37	1.97	1.64	1.27
3	Y	3.20	2.29	1.91	1.61	1.28
	T_∞	3.09	2.33	1.96	1.65	1.28



(see, e.g., Stuart and Ord (18)), a system used to generate a distribution with specific characteristics.

The advantages of the GSH family include: thin tailed distributions can be accommodated; all moments of the distribution are finite; and the calculation of moments of order statistics is more straightforward making data exploring and model fitting simpler. Note also that the GSH behaves like a scaled Laplace or double exponential distribution (see, e.g., Johnson, et al. (13), Chapter 24) in the tails, but does not have a cusp at the mean. These properties and others will be developed in the sequel.

Moments of Order Statistics

Let $Y \sim \text{GSH}(0,1; t)$ with density $g(y)$ and cumulative distribution function $G(y)$. Let Y_r be the r th order statistic of a random sample of size n from Y 's distribution. If $t=0$, the distribution of Y is logistic, and the moments of order statistics can be found in Johnson, et al. (13).

Let $-\pi < t < 0$. Set $u = -t$ and then by transformation,

$$E(Y_r) = \frac{r}{c_2} \binom{n}{r} \sum_{j=0}^{r-1} \binom{r-1}{j} (-1)^j u^{-n+r-j-1} \times \int_0^u [\ln(\sin(u-v)/\sin v)] v^{n-r+j} dv.$$

Using Gradshteyn and Ryzhik (19), 1.518.1 and 1.511, we have

$$\ln \sin v = \ln v + \ln\left(1 - \frac{v}{\pi}\right) + \ln\left(1 + \frac{v}{\pi}\right) - \sum_{j=1}^{\infty} \frac{\zeta(2j) - 1}{j\pi^{2j}} v^{2j}$$

where $\zeta(x)$ is the Riemann Zeta function. Since

$$\begin{aligned} \int_0^u \ln v v^k dv &= \frac{u^{k+1}}{(k+1)^2} [(k+1) \ln u - 1] \\ \int_0^u \ln\left(1 - \frac{v}{\pi}\right) v^k dv &= \frac{\pi^{k+1}}{k+1} \left[\left(\left(\frac{u}{\pi} \right)^{k+1} - 1 \right) \ln\left(1 - \frac{u}{\pi}\right) \right. \\ &\quad \left. - \sum_{j=0}^k \frac{1}{j+1} \left(\frac{u}{\pi} \right)^{j+1} \right] \end{aligned}$$



and

$$\int_0^u \ln\left(1 + \frac{v}{\pi}\right) v^k dv = \frac{\pi^{k+1}}{k+1} \left[\left\{ \left(\frac{u}{\pi}\right)^{k+1} + (-1)^k \right\} \ln\left(1 + \frac{u}{\pi}\right) - (-1)^k \sum_{j=0}^k \frac{(-1)^{j+1}}{j+1} \left(\frac{u}{\pi}\right)^{j+1} \right],$$

using Gradshteyn and Ryzhik (19), 2.729.1, then $\int_0^u \ln \sin v v^k dv$ is expressible as finite sums, known functions and a rapidly converging infinite series. The infinite sum is

$$\pi^{k+1} \sum_{j=1}^{\infty} \frac{\zeta(2j) - 1}{j(2j+k+1)} \left(\frac{u}{\pi}\right)^{2j+k+1}$$

where $u/\pi < 1$ and $\zeta(2j) - 1 = \sum_{h=2}^{\infty} h^{-2j}$. For example, if $j = 15$, then $(\zeta(30) - 1)/15(31+k) \leq 2.003 \times 10^{-12}$. Thus only a few terms are needed for this rapidly converging series, regardless of the value of $t = -u$.

Let $t > 0$. Again after manipulation,

$$E(Y_r) = \frac{r}{c_2} \binom{n}{r} \sum_{j=0}^{r-1} \binom{r-1}{j} \frac{(-1)^j}{t^{n-r+j+1}} \int_0^t [\ln \sinh(t-v) - \ln \sinh v] v^{n-r+j} dv.$$

To evaluate $\int_0^t \ln \sinh v v^k dv$, we write

$$\ln \sinh v = v - \sum_{j=1}^{\infty} \frac{e^{-2jv}}{j} - \ln 2$$

again using Gradshteyn and Ryzhik (19), 1.511. Since

$$\begin{aligned} \int_0^t \sum_{j=1}^{\infty} \frac{e^{-2jv}}{j} v^k dv &= \frac{k!}{2^{k+1}} \zeta(k+2) - \sum_{j=0}^k \frac{k!}{(k-j)!} t^{k-j} \sum_{m=1}^{\infty} m^{-1} (2m)^{-j-1} e^{-2mt} \\ &\equiv I_t(k) \end{aligned}$$

then $\int_0^t \ln \sinh v v^k dv$ is readily evaluated, and hence $E(Y_r)$ is easily computed, even for large sample sizes and regardless of the size of t . In fact, we have:

$$\begin{aligned} \int_0^t [\ln(\sinh(t-v)) - \ln(\sinh(v))] v^k dv &= -\frac{kt^{k+2}}{(k+1)(k+2)} + I_t(k) \\ &\quad - \sum_{j=0}^k \binom{k}{j} (-1)^j I_t(j). \end{aligned}$$



The higher order moments (typically $E(Y_r^2)$ and $E(Y_r Y_s)$ are the only ones needed) are similarly expressed as rapidly convergent series, finite sums and known functions. Note that these formulae do not require special conditions on t , so that we are not restricted in the specific values of kurtosis to be included in the model.

3. PARAMETER ESTIMATION

We assume the shape parameter t is known. In general we say the random variable has a generalized secant hyperbolic distribution, and write $X \sim \text{GSH}(\mu, \sigma; t)$, if its density is given by:

$$f(x) = \frac{c_1}{\sigma} \frac{\exp(c_2(x - \mu)/\sigma)}{\exp(2c_2(x - \mu)/\sigma) + 2a \exp(c_2(x - \mu)/\sigma) + 1} \quad (3.1)$$

where $-\infty < x < \infty$, $\mu \in \mathbf{R}$ and $\sigma > 0$. In order to estimate μ and σ , we take a random sample of size $n \geq 1$, denoted $x_1, x_2, \dots, x_n$, from X 's distribution and define, as usual, the likelihood function L by $L = \prod_{i=1}^n f(x_i)$. To simplify notation, let $h(u) = (a \exp(c_2 u) + 1) / (\exp(2c_2 u) + 2a \exp(c_2 u) + 1)$. Then the likelihood equations for estimating μ and σ are:

$$\frac{\partial \ln L}{\partial \mu} = \frac{c_2 n}{\sigma} - \frac{2c_2}{\sigma} \sum_{i=1}^n h(z_i) = 0 \quad (3.2)$$

and

$$\frac{\partial \ln L}{\partial \sigma} = -\frac{n}{\sigma} + \frac{c_2}{\sigma} \sum_{i=1}^n z_i - \frac{2c_2}{\sigma} \sum_{i=1}^n z_i h(z_i) = 0 \quad (3.3)$$

where $z_i = (x_i - \mu)/\sigma$. It is readily shown that $E(\partial \ln L / \partial \mu) = 0$, $E(\partial \ln L / \partial \sigma) = 0$ and $E(\partial^2 \ln L / \partial \mu \partial \sigma) = 0$. Also, for $-\pi < t < 0$,

$$E\left(\frac{\partial^2 \ln L}{\partial \mu^2}\right) = -\frac{c_2^2 n(t - \sin t \cos t)}{2\sigma^2 t \sin^2 t}, \quad (3.4)$$

and

$$E\left(\frac{\partial^2 \ln L}{\partial \sigma^2}\right) = -\frac{n}{6\sigma^2} \left(\frac{\pi^2 - t^2}{\sin^2 t} - \frac{(\pi^2 - 3t^2) \cos t}{t \sin t} \right) \quad (3.5)$$



while if $t \geq 0$,

$$E\left(\frac{\partial^2 \ln L}{\partial \mu^2}\right) = -\frac{c_2^2 n (\sinh t \cosh t - t)}{2\sigma^2 t \sinh^2 t}, \quad (3.6)$$

and

$$E\left(\frac{\partial^2 \ln L}{\partial \sigma^2}\right) = -\frac{n}{6\sigma^2} \left(\frac{(\pi^2 + 3t^2) \cosh t}{t \sinh t} - \frac{\pi^2 + t^2}{\sinh^2 t} \right). \quad (3.7)$$

The asymptotic variance-covariance matrix is $\mathbf{V} = (V_{ij})$, where $V_{11} = -1/E(\partial^2 \ln L / \partial \mu^2)$, $V_{12} = V_{21} = 0$ and $V_{22} = -1/E(\partial^2 \ln L / \partial \sigma^2)$. Some specific values include:

β_2	t	$n\text{MVB}(\mu)/\sigma^2$	$n\text{MVB}(\sigma)/\sigma^2$
9	$-\pi\sqrt{2/3}$	0.460	1.164
4.2	0	0.912	0.699
42/13	$\pi\sqrt{21/31}$	0.984	0.530
3	π	0.974	0.481
2	$\pi\sqrt{11}$	0.529	0.186

The values of β_2 and hence t were chosen to illustrate the differences between thin and thick tail distributions.

Equations (3.2) and (3.3) do not admit explicit solutions. An iterative process can be used to solve the equations, but without extensive simulations, the properties of the resulting maximum likelihood estimates are difficult to determine, especially for small samples. Other approaches that can be used include Lloyd's (9) Best Linear Unbiased Estimates (BLUEs) and Tiku's robust modified maximum likelihood estimates (MMLEs) (see Tiku (20), (21) and (22)). Tiku's robust estimates are defined in terms of the expected values and variances of order statistics of the standard normal, with a certain amount of censoring imposed. These and other simple estimation schemes require some of the moments of order statistics. For the Student's t family the moments have typically been found by numerical integration methods, and this limits their use for data exploring. As shown earlier, the moments of order statistics for the GSH can be expressed in terms of known functions, finite sums or rapidly converging infinite series. Thus, the GSH family is very amenable for use as a model, since estimates of the parameters and the properties of those estimates are readily available.



Tiku and Suresh (6) introduced modified maximum likelihood estimation in general location-scale models, with the following properties: 1. it is asymptotically equivalent to maximum likelihood when regularity conditions hold (Vaughan and Tiku (23), Bhattacharyya (24) and Tiku et al. (11)); 2. the estimates produced are almost fully efficient in terms of the Minimum Variance Bounds (MVBs) even for small samples; 3. the estimates have little or no bias; 4. the method is essentially self-censoring, since it assigns small weights to extremes; and 5. the estimates are easier to compute than the maximum likelihood estimates and their properties are readily determined. For these reasons, we will use them in our analysis.

Complete sums are invariant to ordering and, with $z_{(i)}$ the i th order statistic of the values $z_1, \dots, z_n$ ($z_i = (x_i - \mu)/\sigma$), $h(z_{(i)}) \simeq h(t_i) + h'(t_i) \times (z_{(i)} - t_i)$. Here and elsewhere, $t_i = E(z_{(i)})$. Hence

$$\frac{\partial \ln L}{\partial \mu} \simeq \frac{\partial \ln L^*}{\partial \mu} = \frac{2c_2}{\sigma} \sum_{i=1}^n \beta_i z_{(i)} = 0 \quad (3.8)$$

and

$$\frac{\partial \ln L}{\partial \sigma} \simeq \frac{\partial \ln L^*}{\partial \sigma} = -\frac{n}{\sigma} + \frac{c_2}{\sigma} \sum_{i=1}^n z_{(i)}(1 - 2\alpha_i + 2\beta_i(z_{(i)} - t_i)) = 0, \quad (3.9)$$

where $\alpha_i = h(t_i)$ and $\beta_i = -h'(t_i)$, $1 \leq i \leq n$. If β_i is negative, we set $\beta_i = 0$. This only occurs when $\beta_2 > 7.5$. Note that $\sum_{i=1}^n \alpha_i = n/2$ and $\sum_{i=1}^n \beta_i t_i = 0$. The simultaneous solutions $\hat{\mu}_m$ and $\hat{\sigma}_m$ of (3.8) and (3.9) are:

$$\hat{\mu}_m = \bar{x}_b = \frac{\sum_{i=1}^n \beta_i x_{(i)}}{\sum_{i=1}^n \beta_i}$$

and

$$\hat{\sigma}_m = \frac{c_2}{2} \left[\bar{x} - \bar{x}_a + \left((\bar{x} - \bar{x}_a)^2 + \frac{8}{nc_2} \left(\sum_{i=1}^n \beta_i x_{(i)}^2 - B \bar{x}_b^2 \right) \right)^{1/2} \right]$$

where $B = \sum_{i=1}^n \beta_i$, $\bar{x} = 1/n \sum_{i=1}^n x_{(i)}$ and $\bar{x}_a = 2/n \sum_{i=1}^n x_{(i)}(\alpha_i + \beta_i t_i)$.

It is readily seen that as $t \rightarrow \infty$ so that the distribution approaches the uniform then $\hat{\mu}_m = (x_{(1)} + x_{(n)})/2$. We will take $\hat{\sigma}_m = (n+1)/(n-1) \times (x_{(n)} - x_{(1)})/(2\sqrt{3})$ here so that the estimates are unbiased and asymptotically identical with the maximum likelihood estimates for the uniform distribution (Johnson et al. (13), p. 286).



4. PROPERTIES OF $\hat{\mu}_m$ AND $\hat{\sigma}_m$

The asymptotic equivalence of the modified likelihood equations with the corresponding likelihood equations and the asymptotic equivalence of the resulting asymptotic variance–covariance matrices are readily established using the results of Hoeffding (25). The asymptotic variance–covariance matrix of $\hat{\mu}_m$ and $\hat{\sigma}_m$, denoted $\mathbf{V}^*$, can be written as

$$\mathbf{V}^* = \sigma^2 \begin{bmatrix} 1/(2c_2 B) & 0 \\ 0 & 1/Q \end{bmatrix}$$

where $B = \sum_{k=1}^n \beta_k$ and $Q = 2c_2 \sum_{k=1}^n (\beta_k t_k^2 + 3\beta_k v_k - 2\alpha_k t_k) - n$. For any fixed value of a , $E(Z^2) < \infty$ ($Z = (X - \mu)/\sigma$) and the functions $h(z)$ and $h'(z)$ are bounded. Thus the conditions of Hoeffding (25), Theorem 1 hold. One consequence is that $\lim_{n \rightarrow \infty} 1/n \sum_{i=1}^n v_i = 0$. Because the β_i s are bounded, $\lim_{n \rightarrow \infty} 1/n \sum_{i=1}^n \beta_i v_i = 0$. Therefore,

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} Q &= \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} E \left(-\frac{\partial^2 \ln L^*}{\partial \sigma^2} \right) = -2c_2 [E(Z^2 h'(Z)) + 2E(Zh(Z))] - 1 \\ &= \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} E \left(-\frac{\partial^2 \ln L}{\partial \sigma^2} \right). \end{aligned}$$

Similar arguments show that $\mathbf{V}$ and $\mathbf{V}^*$ are asymptotically equal, as are the likelihood and modified likelihood equations. $\mathbf{V}^*$ gives better approximations to the variances of the MML estimates for small n than does $\mathbf{V}$.

The rate of convergence of values from the MML estimation method to the corresponding maximum likelihood values is indicated by the following representative results:

n	t	$\sum t_i \alpha_i / (n/2c_2)$	V_{11}^* / V_{11}	V_{22}^* / V_{22}
20	$-\pi/2$	0.961	0.995	0.938
	π	0.907	1.011	0.918
30	$-\pi/2$	0.974	0.996	0.959
	π	0.943	1.010	0.957

Here, for example, V_{ij} is the i, j entry in $\mathbf{V}$. The asymptotic properties can also be inferred from the observation that $1/n = G(t_{i+1}) - G(t_i) + o(1/n) = g(t_i) + o(1/n)$ under general conditions given in Blom (26) and Van Zwet (27).



An alternative might be to use the sample mean $\bar{X}$ and sample standard deviation S . The following results are typical: for samples of size $n = 30$ from $\text{GSH}(0, 1; t)$.

t	$\text{Var}(\bar{X})$	$\text{MVB}(\mu)$	$E(S)$	$\text{Var}(S)$	$\text{MVB}(\sigma)$
$-\pi/4$	0.033	0.0297	0.99	0.0280	0.0241
$-\pi/2$	0.033	0.027	0.984	0.0318	0.0270
$-\pi\sqrt{2/3}$	0.033	0.0153	0.971	0.0570	0.0388

Since iterative methods for generating the maximum likelihood estimates are problematic for thick-tailed distributions, and the MML estimates are almost fully efficient for even relatively small sample sizes, the MML estimates (which are readily computed without iterations) are preferred.

From the above properties and Kendall and Stuart (28), p. 38–53, we have the following results:

Proposition 1. Conditional on σ known, $\hat{\mu}_m$ is (asymptotically) the MVB estimator of μ and asymptotically, $\hat{\mu}_m \sim N(\mu, \sigma^2/(2c_2B))$.

Proposition 2. Conditional on μ known, $\hat{\sigma}_m$ is (asymptotically) the MVB estimator of σ .

Corollary. For large n , we have

$$\frac{\partial \ln L}{\partial \sigma} \simeq \frac{2c_2 \sum_{i=1}^n \beta_i(t_i^2 + v_i)}{\sigma^3} \left(\frac{\sum_{i=1}^n \beta_i(x_{(i)} - \mu)^2}{\sum_{i=1}^n \beta_i(t_i^2 + v_i)} - \sigma^2 \right). \quad (3.15)$$

Therefore, $\hat{\sigma}_m^2(\mu) = \sum \beta_i(x_{(i)} - \mu)^2 / \sum \beta_i(t_i^2 + v_i)$ is MVB for σ^2 and its distribution is a multiple of a chi-square.

5. APPLICATIONS

To illustrate the use of the GSH in analyzing data, we apply the results to two examples, and include some comparative values.

Example 1. Consider the Darwin (29) data on the difference in heights of cross and self fertilised Zea May Plants, as expressed in eighths of



inches in Fisher (30):

Zea May Plants Ordered Differences in Heights in Eighths of Inches							
-67	-48	6	8	14	16	23	24
28	29	41	49	56	60	75	

Fisher's (30) analysis assumed a normal distribution so the usual sample mean and variance were used. The resulting values are:

$$\bar{X} = 20.933, \quad S = 37.74$$

so that, in testing if there is a difference in heights,

$$t_{\text{calc}} = \frac{\sqrt{15}(20.933)}{37.744} = 2.148.$$

Since $t_{\text{crit}}(v = 14, 0.025) = 2.145$ the conclusion is that there is a difference.

An alternative analysis in Tiku et al. (11) uses a robust estimation method. This still assumes normality but censors at least 10% of the data from each tail (Tiku et al. (11), p. 43 and p. 113). In this case that means two observations must be trimmed from each end, i.e., 4/15 of the data. When the four values are trimmed, the robust method yields the following values (reproduced from Tiku et al. (11), p. 43–44):

$$\hat{\mu} = 27.713, \quad \hat{\sigma} = 25.498$$

with respective standard errors

$$\hat{\sigma}_{\hat{\mu}} = 6.743 \quad \text{and} \quad \hat{\sigma}_{\hat{\sigma}} = 5.702.$$

Since $\hat{\mu}/\hat{\sigma}_{\hat{\mu}} \approx 4$ and the estimator is approximately t_{11} the conclusion now is that there is strong evidence of a difference.

Alternatively, $Q-Q$ plots suggest that a GSH with $t = -\pi/2$ gives a good overall fit to the data. A comparison to the normal is given in Figures 1 and 2.

This would correspond to $\beta_2 = 5$. The modified maximum likelihood estimates of μ and σ under the assumption of this value of t are:

$$\hat{\mu}_m = 25.12 \quad \text{and} \quad \hat{\sigma}_m = 35.98.$$



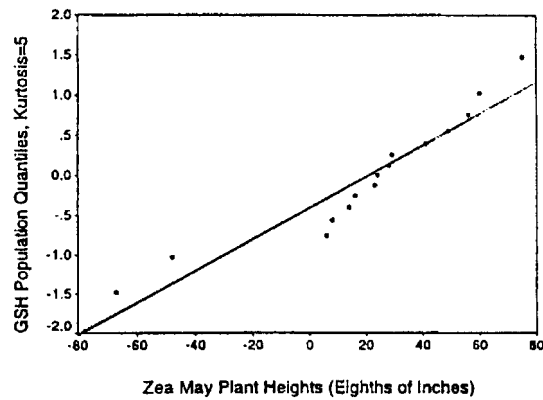


Figure 1.

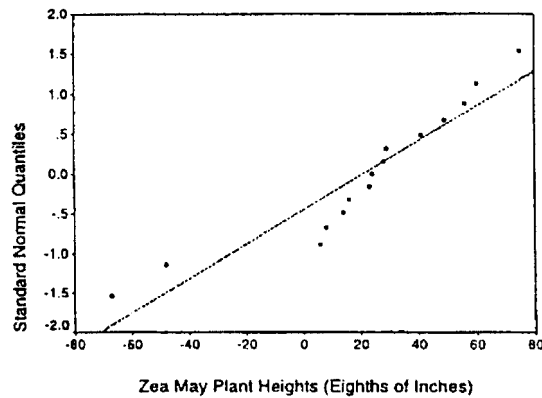


Figure 2.

The estimated standard error needed is:

$$\hat{\sigma}_{\hat{\mu}_m} = 8.39.$$

Asymptotically, $\hat{\mu}_m/\hat{\sigma}_{\hat{\mu}_m}$ is normal, based on Stuart and Ord (18), p. 325, formula 10.17 and the fact that $\hat{\mu}_m$ and $\hat{\sigma}_m$ are asymptotically uncorrelated. For small samples, we can use the approximation to $\hat{\sigma}_m$'s distribution:

$$a \frac{\hat{\sigma}_m^2}{\sigma^2} \sim \chi^2(b).$$



Such approximations have been used effectively in, e.g., Lawless (31), p. 157–159. To find the constants a and b , we use $E(\hat{\sigma}_m/\sigma) \simeq 1$ and $E(U^{1/2}) = \sqrt{2}\Gamma((b+1)/2)/\Gamma(b/2)$, where $U \sim \chi^2(b)$ to show

$$a = 2 \left(\frac{\Gamma((b+1)/2)}{\Gamma(b/2)} \right)^2$$

and b is the solution of

$$2 \left(\frac{\Gamma((b+1)/2)}{\Gamma(b/2)} \right)^2 = \frac{bQ}{Q+1}$$

where Q is as in §4. It suffices, however, to use the asymptotic value $-\sigma^2 E(\partial^2 \ln L / \partial \sigma^2)$ in place of Q . We then have for small samples that $(\hat{\mu}_m - \mu) / \hat{\sigma}_{\hat{\mu}_m} \simeq t(b)$.

Thus, under $H_0: \mu = 0$,

$$P(\hat{\mu}_m / \hat{\sigma}_{\hat{\mu}_m} > 25.12/8.39) \approx 0.008.$$

We also note that the maximum likelihood estimates $\hat{\mu}$ and $\hat{\sigma}$, i.e., the simultaneous solutions of (3.2) and (3.3), are $\hat{\mu} = 25.53$ and $\hat{\sigma} = 35.14$, indicating how close the MML estimates will be to the ML estimates. The ML estimates were found by starting with the MML estimates and iterating the solutions to (3.2) and (3.3). It took one iteration for the estimates to stabilize to 2 decimals of accuracy.

Example 2. The Coronary Heart Disease (CHD) data reported in, e.g., Hosmer and Lemeshow (32) has 100 observations on (X, Y) , where X is the Age (in years) and Y is the presence or absence of coronary heart disease. The main interest in this data is in binary regression of Y on X , where X is taken as a nonstochastic covariate. The analysis assumes that there is a ‘density’ function f (typically logistic) that links X to $P(Y|X = x)$ through

$$E(Y|X = x) = \int_{-\infty}^{\gamma_0 + \gamma_1 x} f(u) du.$$

Q - Q plots assuming the logistic, normal and $\text{GSH}(\mu, \sigma; \sqrt{11}\pi)$, i.e., the GSH member with kurtosis 2 are given in Figures 3–5, respectively. These plots indicate that the GSH with kurtosis 2, gives a very good fit, in terms of straight line regression of $x_{(i)}$ on $t_{(i)}$.



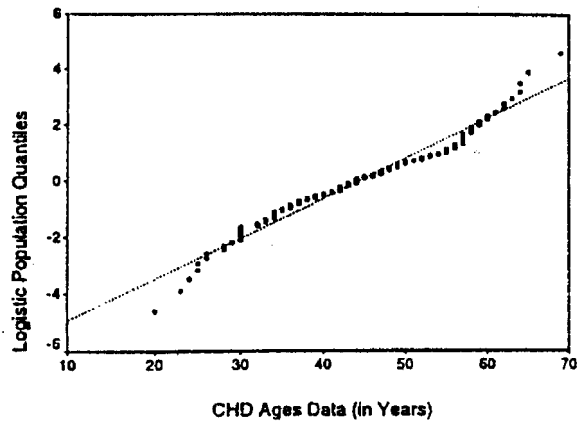


Figure 3.

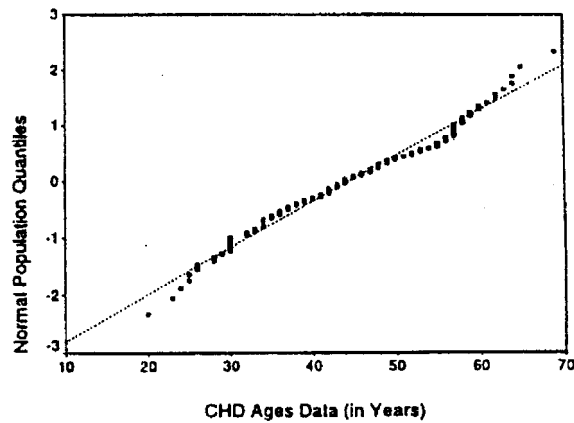


Figure 4.

A comparison of the estimation of the parameters and resulting Wald statistic is given in Table 2.

The Wald (W) statistic is the ratio of the estimate of the parameter to its standard error. See Agresti (33) for details on this and other tests of fit. The MML estimates are based on results in Tiku and Vaughan (34), where logistic and other nonlogistic density functions are considered. These estimates are asymptotically equal to the maximum likelihood estimates, based on arguments similar to those used for the GSH and on Bhattacharyya (24).



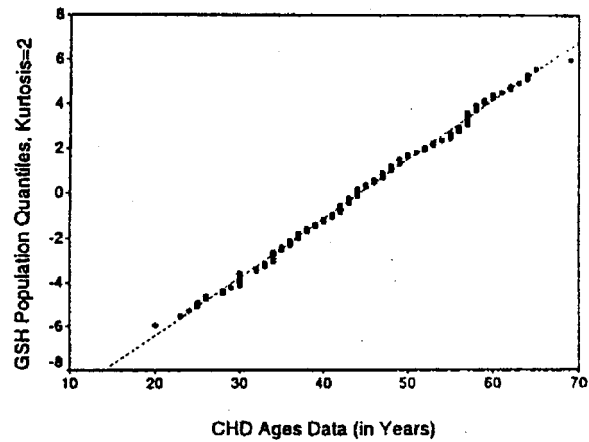


Figure 5.

Table 2. Estimates and Their Standard Errors and the Value of the Wald Statistic (W) for the CHD Data in Example 2

Density	Coefficient	MML Estimate	Standard Error	W
Logistic	γ_0	-2.927	.625	4.59
	γ_1	.0611	.0133	
Normal	γ_0	-3.146	.625	4.91
	γ_1	.0658	.0134	
GSH, $\beta_2=2$	γ_0	-3.296	.507	6.16
	γ_1	.0696	.0113	

6. CONCLUSIONS

Some conclusions that can be drawn from the examples and theory include:

1. the GSH is a viable alternative to Student's t distribution;
2. less trimming of data is necessary when questions about kurtosis and the normal as a model occur;
3. thin tailed distributions can be accommodated;
4. the estimation process is easier from a computational viewpoint: data exploring is simpler since no censoring is required for a well-chosen model; and
5. the fact that all moments are finite should appeal to practitioners.



ACKNOWLEDGMENT

This work was partially supported by the Natural Sciences and Research Council of Canada. The author would also like to thank the Associate Editor for constructive criticisms that have led to an improvement in this work.

REFERENCES

1. Praetz, P.D. The Distribution of Share Price Changes. *J. Business* **1972**, *45*, 49–55.
2. Blattberg, R.C.; Gonedes, N.J. A Comparison of the Stable and Student Distributions as Statistical Models for Stock Prices. *J. Business* **1974**, *47*, 244–280.
3. McLeay, S. Student's *t* and the Distribution of Financial Ratios. *J. Bus. Fin. Accounting* **1986**, *13*, 209–222.
4. Rocke, D.M.; Downs, G.W.; Rocke, A.J. Are Robust Estimators Really Necessary? *Technometrics* **1982**, *24*, 95–101.
5. Spjøtvoll, E.; Aastveit, A.H. Comparison of Robust Estimators on Some Data from Field Experiments. *Scand. J. Statist.* **1980**, *7*, 1–13.
6. Tiku, M.L.; Suresh, R.P. A New Method of Estimation for Location and Scale Parameters. *J. Stat. Plan. Inf.* **1992**, *30*, 281–292.
7. Bian, G.; Tiku, M.L. Bayesian Inference Based on Robust Priors and MML Estimators: Part I, Symmetric Location-Scale Distributions. *Statistics* **1997**, *29*, 317–345.
8. Spjøtvoll, E. Discussion of P. J. Bickel. *Scand. J. Statist.* **1976**, *7*, 164.
9. Lloyd, E.H. Least Squares Estimation of Location and Scale Parameters Using Order Statistics. *Biometrika* **1952**, *39*, 88–95.
10. Barnett, V.D. Order Statistics Estimators of the Location of the Cauchy Distribution. *J. Amer. Statist. Assoc.* **1966**, *61*, 1205–1218. Correction in *63*, 383–385.
11. Tiku, M.L.; Tan, W.-Y.; Balakrishnan, N. *Robust Inference*. Marcel Dekker: New York, 1986.
12. Talacko, J. Perks' Distributions and Their Role in the Theory of Wiener's Stochastic Variables. *Trabajos de Estadística* **1956**, *7*, 159–174.
13. Johnson, N.L.; Kotz, S.; Balakrishnan, N. *Continuous Univariate Distributions*. Vol. 2, 2nd Ed.; John Wiley and Sons: New York, 1995.
14. Perks, W.F. On Some Experiments in the Graduation of Mortality Statistics. *J. Inst. Actuaries* **1932**, *63*, 12–57.



15. Vaughan, D.C. On the Tiku–Suresh Method of Estimation. *Commun. Stat.—Theory Meth.* **1992**, *21*, 451–469.
16. Harkness, W.L.; Harkness, M.L. Generalized Hyperbolic Secant Distributions. *J. Amer. Statist. Assoc.* **1968**, *63*, 329–337.
17. Tiku, M.L.; Jones, P.W. Best Linear Unbiased Estimators for a Distribution Similar to the Logistic. *Proc. Statistics 71 Canada*, Carter, C.S., Dwivedi, T.D., Eds.; 1971; 412–419.
18. Stuart, A.; Ord, J.K. *Kendall's Advanced Theory of Statistics*. Vol. 1, 5th Ed.; Oxford University Press: New York, 1987.
19. Gradshteyn, I.S.; Ryzhik, I.M. *Table of Integrals, Series and Products*. 5th Ed.; Alan Jeffrey, Ed.; Academic Press: 1994.
20. Tiku, M.L. Estimating the Mean and Standard Deviation from a Censored Normal Sample. *Biometrika* **1967**, *54*, 155–165.
21. Tiku, M.L. Estimating the Parameters of Normal and Logistic Distributions from Censored Samples. *Austral. J. Statist.* **1968**, *10*, 64–74.
22. Tiku, M.L. Robustness of MML Estimators Based on Censored Samples and Robust Test Statistics. *J. Stat. Plann. Inf.* **1980**, *4*, 123–143.
23. Vaughan, D.C.; Tiku, M.L. Estimation and Hypothesis Testing for a Non-Normal Bivariate Distribution with Applications. *Mathematical and Computer Modelling—Special Issue on Stochastic Models in Mathematical Biology* **2000**, *32*, 53–67.
24. Bhattacharyya, G.K. The Asymptotics of Maximum Likelihood and Related Estimators Based on Type II Censored Data. *J. Amer. Statist. Assoc.* **1985**, *80*, 398–404.
25. Hoeffding, W. On the Distribution of the Expected Values of the Order Statistics. *Ann. Math. Stat.* **1953**, *24*, 93–100.
26. Blom, G. *Statistical Estimates and Transformed Beta-Variables*. Uppsala: Almqvist and Wiksells, 1958.
27. Van Zwet, W.R. *Convex Transformations of Random Variables*. Amsterdam: Mathematical Centre Tracts 7, Mathematisch Centrum, 1964.
28. Kendall, M.; Stuart, A. *The Advanced Theory of Statistics*. Vol. 2, 4th Ed.; Charles Griffin: London, 1979.
29. Darwin, C. *The Effects of Cross- and Self-Fertilization in the Vegetable Kingdom*. John Murray: London, 1876.
30. Fisher, R.A. *The Design of Experiments*. 8th Ed.; Hafner: New York, 1971.
31. Lawless, J.F. *Statistical Models and Methods for Lifetime Data*. John Wiley: New York, 1982.



32. Hosmer, D.W.; Lemeshow, S. *Applied Logistic Regression*. John Wiley and Sons: New York, 1989.
33. Agresti, A. *An Introduction to Categorical Data Analysis*. John Wiley and Sons: New York, 1996.
34. Tiku, M.L.; Vaughan, D.C. Logistic and Nonlogistic Density Functions in Binary Regression with Nonstochastic Covariates. *Biometrical J.* **1997**, *39*, 883–898.



Request Permission or Order Reprints Instantly!

Interested in copying and sharing this article? In most cases, U.S. Copyright Law requires that you get permission from the article's rightsholder before using copyrighted content.

All information and materials found in this article, including but not limited to text, trademarks, patents, logos, graphics and images (the "Materials"), are the copyrighted works and other forms of intellectual property of Marcel Dekker, Inc., or its licensors. All rights not expressly granted are reserved.

Get permission to lawfully reproduce and distribute the Materials or order reprints quickly and painlessly. Simply click on the "Request Permission/Reprints Here" link below and follow the instructions. Visit the [U.S. Copyright Office](#) for information on Fair Use limitations of U.S. copyright law. Please refer to The Association of American Publishers' (AAP) website for guidelines on [Fair Use in the Classroom](#).

The Materials are for your personal use only and cannot be reformatted, reposted, resold or distributed by electronic means or otherwise without permission from Marcel Dekker, Inc. Marcel Dekker, Inc. grants you the limited right to display the Materials only on your personal computer or personal wireless device, and to copy and download single copies of such Materials provided that any copyright, trademark or other notice appearing on such Materials is also retained by, displayed, copied or downloaded as part of the Materials and is not removed or obscured, and provided you do not edit, modify, alter or enhance the Materials. Please refer to our [Website User Agreement](#) for more details.

[Order now!](#)

Reprints of this article can also be ordered at

<http://www.dekker.com/servlet/product/DOI/101081STA120002647>